

Solution

HUMAN EYE -05

Class 10 - Science

Section A

1. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.

2. (a) Both A and R are true and R is the correct explanation of A.

Explanation: A person with myopia can see nearby objects clearly but cannot see distant objects distinctly. A person with this defect may see clearly up to a distance of a few metres. In a myopic eye, the image of a distant object is formed in front of the retina but not at the retina itself. This defect may arise due to excessive curvature of the eye lens. Thus, both assertion and reason are true and reason is the correct explanation of assertion.

- 3.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: The refractive defects can be corrected by contact lenses and eye lens is composed of fibrous jelly-like material. Thus, both assertion and reason are true but the reason is not the correct explanation of assertion.

- 4.

- (c) A is true but R is false.

Explanation: Sometimes, the crystalline lens of people at old age becomes milky and cloudy. This condition is called cataract. This causes partial or complete loss of vision. The eyeball is spherical in shape and not elliptical. Thus, the given assertion is true and reason is false.

- 5.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: The person cannot see the objects distinctly and comfortably because of accommodation. The vision becomes blurred due to the refractive defects of the eye. Therefore, the given assertion is true, but reason is not the correct explanation of assertion.

6. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.

7. (a) Both A and R are true and R is the correct explanation of A.

Explanation: For thin prism, the angle of prisms A is small. For small A, $D_{\min}$ (minimum deviation) is also small.

$$\text{So, } \mu = \frac{\sin\left(\frac{A+D_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)} \dots (i)$$

$$\sin\left(\frac{A+D_{\min}}{2}\right) \approx \frac{A+D_{\min}}{2} \quad (\sin\theta \approx \theta \text{ for small } \theta)$$

$$\text{and } \sin \frac{A}{2} \approx \frac{A}{2}$$

Using above approximation, in equation (i)

$$\mu = \frac{\frac{A+D_{\min}}{2}}{\frac{A}{2}}$$

$$D_{\min} = (\mu - 1)A$$

Hence, it can be seen that if A is small, $D_{\min}$ is also small.

- 8.

- (d) A is false but R is true.

Explanation: Higher the refractive index of the prism material, greater is the angle of deviation.

9. (a) Both A and R are true and R is the correct explanation of A.

Explanation: In a vacuum speed of light is independent of wavelength, Hence, no dispersion takes place in a vacuum. Thus, a vacuum is a non-dispersive medium in which all colors travel at the same speed.

10. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.

11. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
12. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The continuously changing atmosphere is able to cause variation in the light coming from a point-sized star because of which the star appears to be twinkling.
13. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: As planets are of larger size than stars and much closer to the earth, planets can be considered as a collection of a large number of point-sized sources of light. The total variation in the amount of light entering our eye from all these individual point-sized sources will average out to zero which nullifies the twinkling effect of each other. Therefore, planets do not twinkle.
14. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: During the daytime, the sky appears blue. This is because the size of the particles in the atmosphere is smaller than the wavelength of visible light, so they scatter the light of shorter wavelengths. The scatter blue light enters our eyes.
15. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
16. **(c)** A is true but R is false.
Explanation: Red colour has longest wavelength.
17. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
18. **(c)** A is true but R is false.
Explanation: A is true but R is false.
19. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: We can see the sun and its surroundings appear red colour during both sunrise and sunset. It happens due to the scattering of light is inversely proportional to the fourth power of its wavelength. Thus, both assertion and reason are true and reason is the correct explanation of assertion.
20. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
21. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
22. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
23. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
24. **(c)** A is true but R is false.
Explanation: A is true but R is false.
25. **(d)** A is false but R is true.
Explanation: Magnification produced by mirror,

$$m = \frac{I}{O} = \frac{f}{f-u} = \frac{f}{x}$$
Where, x is distance from focus.
and $m = \frac{\text{Size of image (I)}}{\text{Size of object (O)}}$

26. (a) Both A and R are true and R is the correct explanation of A.

Explanation: $RP \propto \text{diameter of objective}$

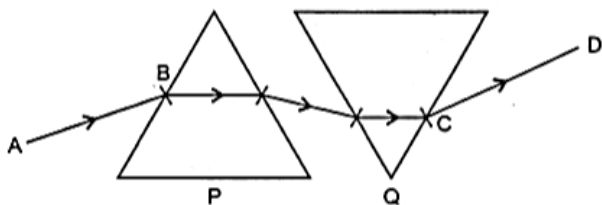
Section B

27. (a) All the colours of the white light move with the same speed

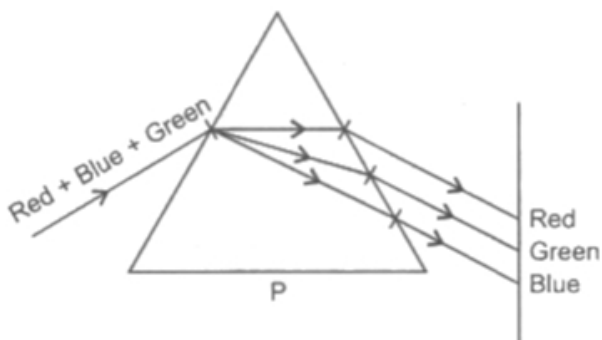
Explanation: All the colors of white light move with the same speed because speed of light doesn't depend on color and has a constant value.

Section C

28. The completed diagram is as follows in which the ray CD which is emerging out of the prism Q is parallel to the ray AB which is incident on the prism P.



29. The completed ray diagram shows the dispersion of (Red + Blue + Green) light by the prism.



30. **Glass slab:** It is rectangular in shape and made up of glass. In glass slab, the direction of the incident ray and the emergent ray of light are parallel to each other.

Glass Prism: It has two triangular sides, two inclined rectangular sides, and one rectangular base. In glass prism, the direction of the incident ray and the emergent ray of light is not parallel to each other.

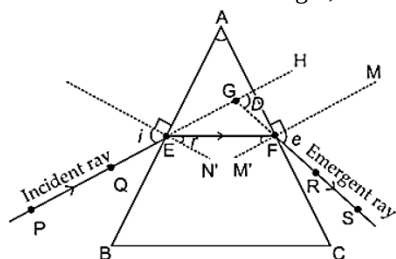
When a narrow beam of monochromatic light passes through:

- Glass slab: It deviates from the actual path but the direction of the incident ray and the emergent ray of light are parallel to each other.
- Glass prism: It has deviated from the actual path but the direction of the incident ray and the emergent ray of light are not parallel to each other.

When a narrow beam of white light passes through:

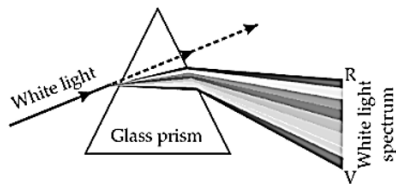
- Glass slab: The splitting of white light into its constituent color does not occur. And the direction of the incident ray and the emergent ray of light are parallel to each other.
- Glass prism: The splitting of white light into its constituent seven colors occurs. And the direction of the incident ray and the emergent ray of light is not parallel to each other.

31. i. In case of monochromatic light, the ray will bend due to refraction but it will not show the dispersion of light.

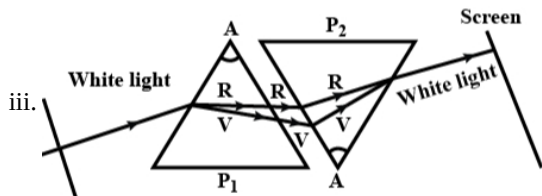


- ii. If narrow beam of monochromatic light is replaced with a beam of white light, a combination of narrow beam of seven colour will come out. The phenomenon of splitting of white light into its seven constituent colours when it passes through a glass prism is called dispersion of white light. The various colours seen are Violet, Indigo, Blue, Green, Yellow, Orange and Red.

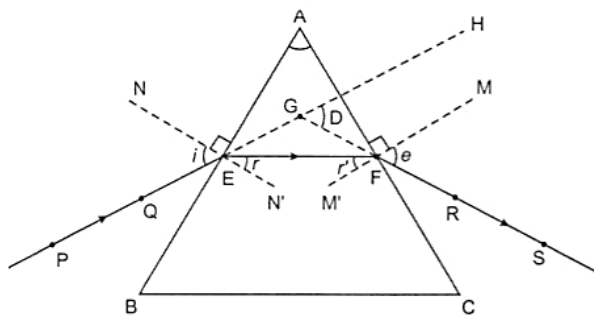
The different component colour of light bends at a different angle w.r.t the incident angle.



32. i. The relation between colour of scattered light and size of the scattering particle is Small size particles scatter shorter wavelength (violet) or large sized particles scatter larger wavelength (Red).
 ii. The basic cause of this observation is due to variation in physical condition of hot air.

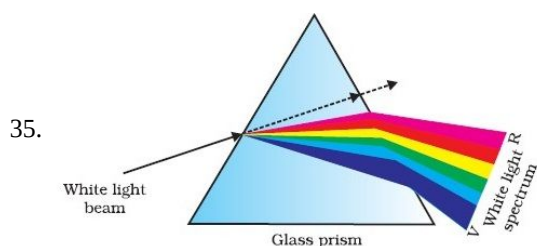


33. The labelled diagram has been shown in figure given below, in which



PE → Incident ray
 $\angle i$ → Angle of incidence
 EF → Refracted ray
 $\angle r$ → Angle of refraction
 FS → Emergent ray
 $\angle e$ → Angle of emergence
 $\angle A$ → Angle of the prism
 $\angle D$ → Angle of deviation

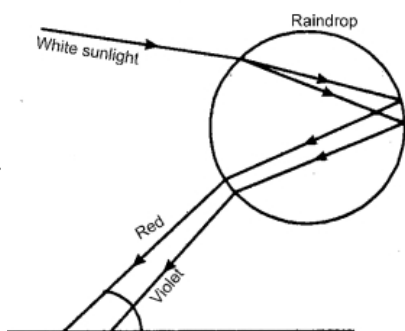
34. i. Different colours of light bend through different angles with respect to the incident ray as they travel with different speeds while passing through a prism, this phenomenon is known as dispersion of light.
 ii. X = violet, this colour has minimum wavelength thus suffers maximum deviation, Y = red, as it has maximum wavelength and thus least deviated.



Order of colour from bottom to top: Violet, Indigo, Blue, Green, Yellow, Orange and Red.

36. Light rays can split into its constituent colours by phenomenon of dispersion where different colours travel with the different speeds in any medium other than air and vacuum and bend through the different angles and split.
 Newton showed that the reverse of dispersion of light is also possible. He kept two prisms close to each other one in erect position and the other in an inverted position. The dispersion of light occurs when it passes through the first prism. The second prism receives all the seven colored rays from first prism and recombines them into the original white light. This observation shows that sunlight is made up of seven colours.
 Any light that gives spectrum similar to that of sunlight is called white light.

37.



Rainbow is caused by dispersion of white sunlight by tiny water droplets present in the atmosphere. Water droplets act as tiny prisms. They refract and disperse the incident sunlight, then reflect it internally and finally refract it again when it comes out of the raindrop. Rainbow is always formed in a direction opposite to that of the sun.

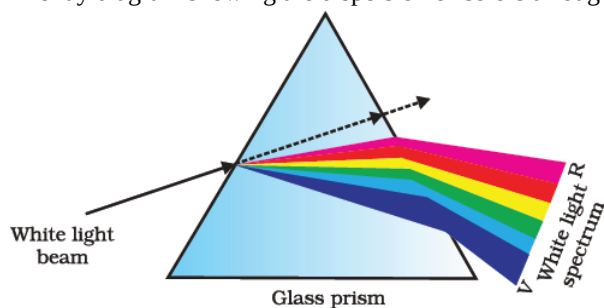
38. Rainbow is always formed in a direction opposite to that of the Sun after rainfall. Where small water droplets present in the atmosphere act like small prisms. They refract and disperse the incident light, come out of the drop and split the sunlight into its constituent colours. Due to this, different colours reach observer's eyes.

39. Dispersion is the phenomenon of splitting of white light into seven colors.

The band of colored components of seven colors is known as a spectrum.

As light ray travels through a glass prism, different colors of light bend at different angles to the incident ray. The violet light bends the most and the red light bends at least angle. The rays of prism bend along different paths and become distinct.

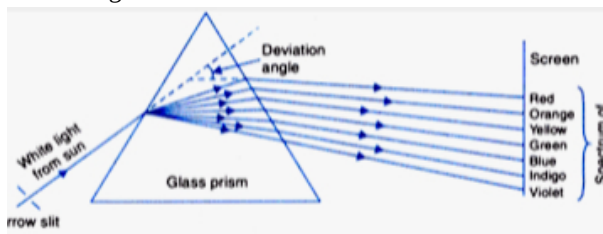
The ray diagram showing the dispersion of colors through a prism can be shown as follows:



40. i. No the statement made by the student is incorrect as because 3 refers to yellow which is the colour of gold metal and 5 refers to blue colours of the spectrum which is the colour of sky.

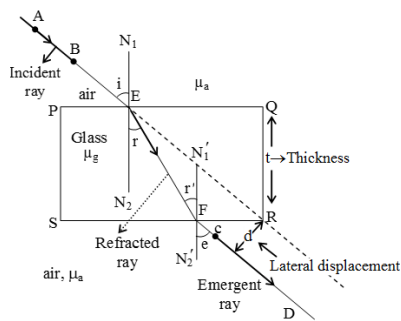
- ii. a. 7 which is violet
- b. 1 which is red
- c. 6 which is indigo
- d. 2 which is orange

41. The splitting up of white light into its constituent colours on passing through a refracting medium like a glass prism is called dispersion of light. The dispersion of white light occurs because different colours of light bend through different angles with respect to the incident ray, as they pass through a prism. Thus, each colour emerges along a different path, forming a spectrum. The red light bends the least while the violet the most as shown below.

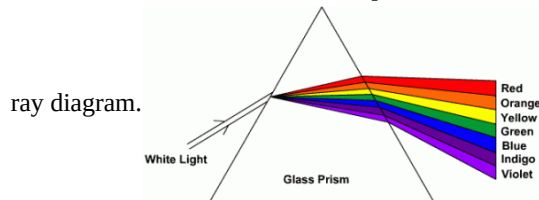


42. The phenomenon of splitting up of white light into its constituent colours as it passes through prism is known as dispersion. Light rays bend towards the normal when it enters the prism. Since white light consists of 7 colours it splits into 7 bands of colour.

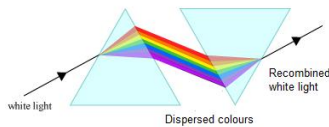
43. a. i. The incident beam of light after refraction through glass slab emerges out parallel to the incident beam but laterally shifted. But no dispersion takes place in this case. The refraction through glass slab is shown in the ray diagram.



- ii. The incident beam of white light after refraction through a prism splits into a band of seven colors which are violet, indigo, blue, green, yellow, orange and red. These colored rays emerge out of the prism along different directions and become distinct. Therefore, the dispersion of white light takes place. Dispersion of white light through prism is shown in

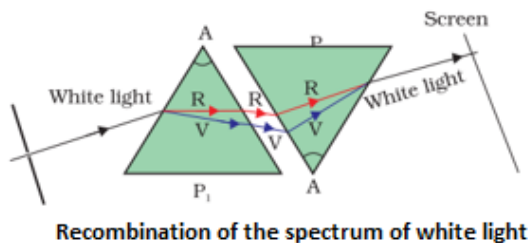


- iii. When the incident beam passes through the first prism, it gets split into the band of seven colors. But those colored rays are incident on an identical inverted prism. Then recombination of the colored rays takes place. This emergent light is parallel to the incident beam but slightly shifted outward. It can be represented with ray diagram as follows:

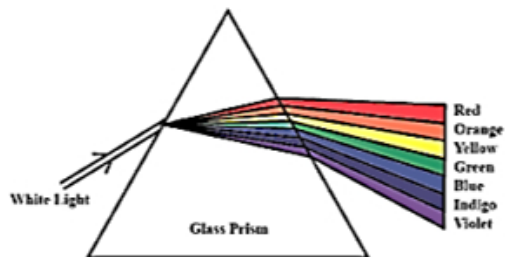


- b. As, It is clear from ray diagram that in case (i) and (iii), the beams of emergent ray is parallel to the incident beam and are slightly shifted. Therefore, emergent beam in the case (i) and (iii) are similar.

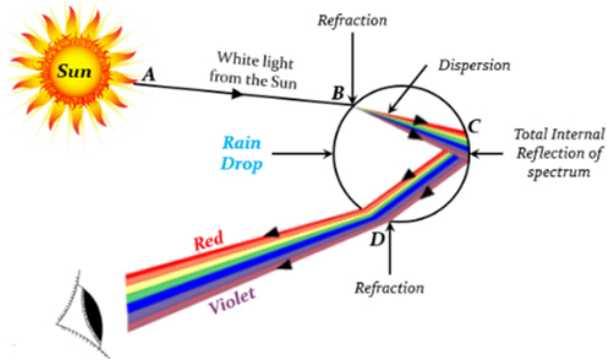
44. For this experiment, one prism is placed near another prism so that one prism is in erect position that is base placed downwards and another one is in inverted position (apex downwards) as shown in the diagram below. When ray of white light enters the first prism, dispersion of light takes place into a 7 colour band and When lights of different colours pass through the second prism, they recombine to make a ray of white light.



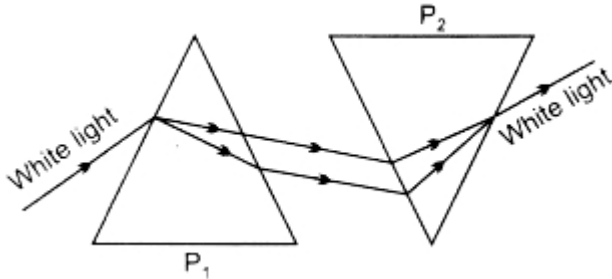
45. i. Dispersion of a beam of white light through glass prism



ii. Formation of a rainbow.



46. When white light is refracted through a glass prism, it gets split into its constituting colours at different angles. This phenomenon is called Dispersion of Light. Forming a rainbow,



Least deviated colour is red whereas most deviated colour is violet. When second identical prism is placed in an inverted position with respect to first prism, recombination of the spectrum will take place and white light will be obtained.

47. We see a rainbow in the sky only after rainfall due to the dispersion of the sunlight by the tiny water droplets of rain suspended in the atmosphere. The water droplets act as a small prism. The raindrops are refracted, dispersed, and reflected internally and then refracted again. Due to these phenomena, we see a rainbow in the sky after rainfall.

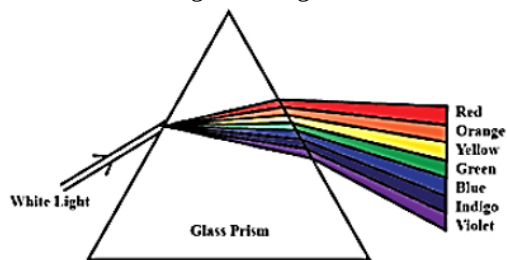
48. Green, yellow, orange and red light have wavelength longer than blue light.

49. The diagram depicts the phenomenon of dispersion of light. Label A indicates incident white light. Label B indicates spectrum consisting of seven colours (VIBGYOR).

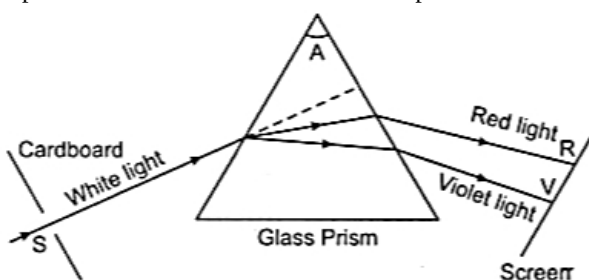
50. The splitting of a beam of white light into its seven constituent colors, when it passes through a glass prism, is called the dispersion of light.

i. Violet is the colour that bends the most because it has the shortest wavelength and is refracted the most.

ii. Red, on the other hand, is the colour that bends the least because it has the longest wavelength and experiences the least amount of refractive error. Indigo, blue, green, yellow, and orange are the other colours of the spectrum, which are indicated in order of increasing wavelength.



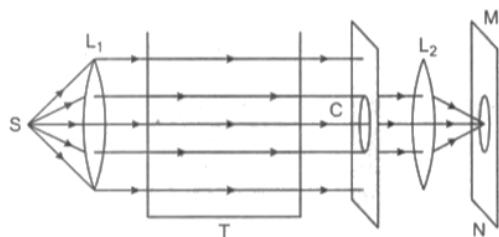
51. When a beam of white light passes through a glass prism it splits up into its constituent seven colours. The splitting of white light into its constituent colours when light passes through a dispersive medium is called dispersion of light. The seven colours, usually expressed as VIBGYOR constitute the spectrum of white light.



The ray diagram showing dispersion is given in fig.

52. The clouds contain dust particles and water molecules of size bigger than the wavelength of visible light, so they scatter all colours of incident white light from sun to the same extent and hence when the scattered light reaches our eyes, the clouds are seen white.
53. i. Atmospheric refraction is responsible for advanced sunrise.
 ii. Scattering of light is responsible colour of water in deep sea.
 iii. Atmospheric refraction causes twinkling of stars.
 iv. Scattering of light leads to reddish appearance of sky during sunset and sunrise.
54. As Planets can be considered as a collection of a large number of point-size sources of light, Planets do not twinkle because they appear larger in size than the stars as they are relatively closer to earth. The different parts of these planets produce either brighter or dimmer effect in such a way that the average of brighter and dimmer effect is zero. Hence, the twinkling effects of the planets are nullified and they do not twinkle.
55. Stars twinkle due to atmospheric refraction of starlight. The starlight passes through atmospheric medium of gradually changing refractive index. So, it is refracted multiple times. As the apparent position of star and the physical condition of the earth's atmosphere are not constant the amount of starlight entering the eye flickers and gives a twinkling effect.
56. No, the position of a star as seen by us is not its true position that is the apparent position. The stars appear slightly higher than the actual position because the light coming from the stars suffers refractions due to different layers of atmosphere which is due to different refractive indices value. It seems as if stars are slightly higher than they actually appear.
57. i. Danger signals are red in color because it is scattered the least by air molecules. The effect of scattering is inversely related to the fourth power of the wavelength of a color. Since the color red has the highest wavelength of all the colors we can see it scattered the least. So, red light is used as a danger signal as it is able to travel the longest distance through the fog, rain, etc.
 ii. In outer space, the sky appears dark instead of blue to an astronaut because there is no atmosphere containing air in outer space to scatter sunlight. Since there is no scattered light to reach our eyes in outer space, the sky appears to be dark there.
 iii. Due to atmospheric refraction, the sun is visible two minutes after the actual sunset. Thus, the time difference between the actual sunset and apparent sunset is 2 minutes.
58. In the forest, mist contains tiny droplets of water which act as particles of colloid dispersed in air. When the sunlight enters the dense forest, the rays of light pass through these particles of colloids and get scattered. Due to the scattering of light, the path of the sun ray is clearly visible. This is called the 'Tyndall effect'.
59. The atmosphere exists only upto some height. Above that there is no atmosphere to scatter the light. There is no scattering of light. The sky therefore appears to be perfectly black to astronauts.
60. Blue colour of the sky as seen from earth surface is on the account of scattering of light of shorter wavelength by particles present in the atmosphere of the earth. If the earth had no atmosphere, there would not have been any scattering and sky would have looked dark. When an astronaut is in his spacecraft, he is present above the atmosphere of earth, from where the sky appears dark to him because there is no scattering of light there.
61. Before reaching our eyes the sunlight near the horizon travels through → **thick layers** of air and → larger distance in earth atmosphere. The **overhead Sun** → travels a shorter distance. Since little of the blue and violet are → scattered the noon Sun appears white. Near the horizon, most of the blue light and shorter wavelength are → scattered by the particles. The Sun **reddish appearance** is because → of the light reaches our eyes is of shorter wavelength.
62. At noon, the sun is overhead and the light coming from the sun travels a relatively shorter distance through the atmosphere to reach the earth. In this case the blue light is not scattered much. As the light coming from the overhead sun contains almost all its component of colours in the right proportion, the sun appears white to us at noon.
63. Our eyes have rods and cones, which help us to see the object and distinguish the colour. Rods are more sensitive to the light but unable to differentiate the colour. Cones are less sensitive to the light but able to differentiate the colour. In dim light only rods are active. So, we are able to see the objects but can not distinguish between colours.
64. a. The danger signals installed at the top of tall buildings are red in colour. This is because red light has the longest wavelength and hence is scattered least by fog or smoke.
 b. The scattering of the light takes place because of the particles present in the atmosphere. At high altitudes, due to the absence of the atmosphere, the scattering of light does not take place. Hence, the sky appears dark to the passengers flying at high altitudes.
 c. When a beam of light is passed through a colloidal solution, then a scattering of light is observed. This is known as the Tyndall effect. This scattering of light illuminates the path of the beam in the colloidal solution.
65. At noon, the sun is directly above our head and we get the light rays directly from the sun after travelling the shortest distance, without much scattering of any particular colour. Hence, the sky appears white.

66. (a) It is incorrect as length of the day in space would be about four minutes shorter than that on the earth.
 (b) It is correct because in space we cannot observe scattering of light because of absence of atmosphere.
 (c) It is incorrect because twinkling of stars is due to atmospheric refraction but in space we do not have atmosphere and atmospheric particles.
67. I. An arrangement for observing the scattering of light in the laboratory is as shown below.



- II. i. On the screen, first orange red colours and then bright crimson red colour patch is observed.
 ii. From the other three sides of colloidal solution of sulphur in a glass tank (T), blue colour is observed.
- This is because the very fine colloidal sulphur particles scatter away the blue colour from the path of beam and only red colour (least scattered) of the beam of white light reaches the screen through the solution.
68. Since, scattering $\propto d$, where d is a diameter of the particle also, scattering $\propto \frac{1}{\text{Wavelength}}$.
- The colour of the scattered light depends on the size of the scattering particles. If the size of the particles is very fine, they scatter mainly light of shorter wavelength or blue light. If the size of particles is larger, they scatter light of longer wavelength (light of red and orange colour). If the size of particles is large enough, they scatter light of all colours and appear white.
69. Sun at noon is at overhead and travels a shorter distance and is therefore white in color.
- During sunrise/sunset, light from the Sun near the horizon passes through different thicker layers of air and larger distance is traveled. So, the color of the Sun is reddish because of the scattering of light, and since Red color has the highest wavelength and is scattered least.
70. Fog affect the visibility of an area. On a foggy day, most of the light gets scattered by the particles in the fog and the visibility also reduces. So it is difficult to drive on a foggy day.
71. The colour of an emergent light when white light is passed through a colloidal solution of sulphur is orange red or bright crimson.
- It is because the minute colloidal particles sulphur scatter shorter wavelength, so the emergent light is of larger wavelength.
72. Scattering of light is the phenomenon in which a part of the incident light is dispersed in different directions.
- Dependence of colour and scattered light on the size of particles:
- When the particles like dust and water droplets present in the atmosphere are large in size, the scattered light appears white.
 - When the particles are extremely minute in size, they will scatter blue light present in the white sunlight.
73. In addition to the exercise for the strained eyes, my medical suggestion would be the usage of the anti-glare lens in their spectacles which reduces the impact of blue light emitted from digital items and artificial lighting. This would reduce the stress and strain caused. But still the exercise for eyes in repeated time interval should also be regularly followed.